System Requirements Specification - Anti-Front-Running ZK Perpetuals on Cardano using Midnight

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1. Introduction

1.1 Purpose

This document specifies the system requirements for a **privacy-preserving perpetual derivatives trading protocol built using Cardano and the Midnight network**.

The system introduces **private order flow and verifiable settlement using zero-knowledge proofs**, preventing front-running and MEV exploitation in decentralized derivatives markets.

The protocol leverages:

- Midnight confidential smart contracts
- zero-knowledge proof verification
- selective disclosure mechanisms
- private state transitions verified through cryptographic proofs

Zero-knowledge proofs allow a participant to prove the correctness of a statement **without revealing the underlying private data**, enabling confidential financial operations on a public blockchain.

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2. System Overview

2.1 Problem Statement

On most public blockchains, transactions are visible in the mempool before confirmation.

This transparency allows adversaries to:

- observe pending trades
- insert competing transactions
- reorder execution
- extract profit through MEV strategies

These attacks cause:

- unfair execution
- strategy leakage
- slippage manipulation
- loss of capital for legitimate traders

Perpetual derivatives trading is particularly vulnerable because **execution order and timing directly influence profitability**.

2.2 Solution Overview

The system introduces a **confidential trading infrastructure using Midnight's privacy framework**.

Key principles:

1. **Private transaction execution**
2. **Selective disclosure of data**
3. **ZK-verified state transitions**
4. **Proof-based settlement verification**

Midnight enables applications to prove transaction validity **without exposing sensitive data**, providing programmable privacy for decentralized applications.

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3. Functional Requirements

3.1 Private Order Submission

The system must allow traders to submit perpetual trading orders including:

- trading pair
- order direction (long / short)
- leverage
- margin allocation
- price
- order size

Order parameters must remain **confidential and protected by cryptographic commitments**.

3.2 Order Commitment

Orders must be represented as **commitments generated locally by the trader**.

Each commitment must contain:

- hashed order parameters
- trader signature
- nonce value

This prevents adversaries from inspecting trade parameters prior to execution.

3.3 Proof-Based Order Validation

Before an order can be processed, the system must verify a **zero-knowledge proof demonstrating**:

- the trader owns sufficient collateral
- the order format is valid
- margin requirements are satisfied

The verification must occur **without revealing the private values used in the proof**.

3.4 Confidential Trade Matching

Orders are matched according to protocol rules:

- price compatibility
- liquidity availability
- margin constraints

Matching logic may occur:

- within private execution environments
- through cryptographic verification circuits

Correct matching must be verified through proof generation before settlement.

3.5 Settlement

Once a match occurs, the system must update:

- trader balances
- margin accounts
- open positions

State transitions must be verified using **zero-knowledge proofs before being accepted by the network**.

3.6 Liquidation Engine

The protocol must support liquidation when margin requirements are violated.

Liquidation must verify:

- margin ratios
- position exposure
- liquidation penalties

The liquidation decision must be validated using a cryptographic proof to prevent manipulation.

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4. Non-Functional Requirements

4.1 Privacy

The protocol must protect the confidentiality of:

- order parameters
- trader identities
- margin balances
- open positions

4.2 Security

The protocol must prevent:

- front-running
- transaction replay attacks
- order tampering
- malicious validator manipulation

4.3 Scalability

The architecture must support high-frequency trading workloads without significant latency increases.

4.4 Reliability

The system must ensure correct settlement even during high network usage or adversarial conditions.

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5. System Architecture

5.1 High Level Architecture

```
Trader Interface
       |
Local Order Construction
       |
Order Commitment
       |
Private Execution Layer
       |
Proof Generation
       |
Midnight Smart Contract Verification
       |
Cardano Anchored Settlement
```

5.2 Core System Modules

zk-circuits

Responsible for generating and verifying cryptographic proofs for:

- order validation
- matching correctness
- settlement updates
- liquidation conditions

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matching

Handles:

- order intake
- matching logic
- liquidity coordination

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settlement

Handles:

- margin updates
- position management
- liquidation execution

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privacy

Implements:

- encryption
- commitment generation
- identity shielding

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common

Contains shared components including:

- cryptographic utilities
- protocol configuration
- data structures

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6. Midnight Integration

Midnight provides the **confidential execution environment** for the protocol.

The architecture leverages three main Midnight components.

6.1 Kachina Execution Framework

The protocol uses the **Kachina framework**, which allows users to maintain private state locally and submit **zero-knowledge proofs demonstrating valid state transitions**.

Only the proof is submitted to the blockchain, ensuring sensitive data remains private.

6.2 Compact Smart Contract Language

Smart contracts are written using **Compact**, Midnight's smart contract language designed with privacy as the default behavior.

Data must be explicitly disclosed, preventing accidental leakage of sensitive information.

6.3 Proof Server

Proof generation is performed using a **local proof server**, which generates cryptographic proofs validating the correctness of private computations before submission to the network.

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7. Zero-Knowledge Proof Design

The system relies on **non-interactive zero-knowledge proofs**, allowing a prover to demonstrate correctness of computations without revealing private inputs.

This enables confidential financial transactions while maintaining network verifiability.

Primary proof circuits include:

Order Validation Circuit

Validates:

- signature correctness
- margin sufficiency

- order integrity

Matching Verification Circuit

Validates:

- correct order pairing
- fair execution
- liquidity availability

Settlement Verification Circuit

Validates:

- balance updates
- position changes
- margin accounting

Liquidation Verification Circuit

Validates:

- margin breach conditions
- liquidation eligibility
- penalty application

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8. Performance Benchmarks

The following performance benchmarks define target system capabilities.

Metric	Target
Proof generation time	≤ 200 seconds
Proof verification time	≤ 8 minutes
Settlement throughput	≥ 5 trades per second
Order commitment latency	≤ 2 seconds

These values are **engineering estimates** that will be validated during development.

9. Privacy Guarantees

The protocol guarantees the following privacy properties.

9.1 Order Confidentiality

Trade parameters remain encrypted and are never revealed to external observers.

9.2 Participant Anonymity

Trader identities remain protected through cryptographic commitments and shielded execution.

9.3 Strategy Protection

Observers cannot infer trading strategies from:

- order flow
- settlement transactions
- state updates

10. Threat Model

10.1 Adversaries

Potential adversaries include:

- MEV bots
- malicious traders
- compromised nodes
- external attackers

10.2 Cryptographic Assumptions

Security relies on:

- correctness of zero-knowledge proof verification
- collision resistance of hash functions
- secure digital signature schemes
- integrity of Midnight's execution framework

11. Attack Vectors and Mitigations

Attack	Description	Mitigation
Front-running	Bots inspect pending transactions	Private order commitments
Replay attacks	Reusing previously signed orders	Nonce verification
Order manipulation	Tampering with order parameters	ZK proof validation
Validator collusion	Manipulating execution order	Confidential execution
State forgery	Invalid settlement state	Proof-verified transitions

12. Risk Assessment

Technical Risks

Risk	Impact
inefficient proof generation	increased latency
circuit complexity	reduced scalability
cryptographic vulnerabilities	protocol compromise

Mitigation Strategies

- incremental circuit optimization

- batch proof aggregation
- formal security audits
- staged testnet deployment

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13. Repository Structure

The public repository will contain the following modules:

```
root
│
├── zk-circuits
├── matching
├── settlement
├── privacy
├── common
└── docs
      └── SRS.md
```